

# IdeaGen: Agentic Multi-Model Decision Platform

Project One-Pager | Portfolio artifact for AI engineering roles

## Problem

Most AI ideation tools produce one-shot text, but business teams need decision-grade workflows: cross-model evaluation, run-to-run comparisons, saved evidence, and stakeholder-ready outputs.

## Key Features

- Multi-provider generation: OpenAI, Gemini, DeepSeek, Grok
- Per-run ranking with a structured quality rubric
- Run A vs Run B comparison with winner + tradeoff rationale
- Decision Summary report across selected runs (1-5)
- Execution Plan (Investment Readiness Assessment)
- PDF export + email delivery, plus saved artifact lifecycle
- Usage controls: auth, quotas, plan gating, storage caps

## Architecture

FastAPI acts as the deterministic control plane, while task-specific agents handle generation, ranking, comparison, reporting, recommendations, and email composition.

UI (Next.js) -> FastAPI Orchestrator -> Agent Modules -> LLM Providers + SQLite: saved results/reports + usage tracking + WeasyPrint: HTML-to-PDF report rendering + Resend: transactional email delivery + Clerk JWT + JWKS: identity and API protection

## Technology Stack

**Frontend:** Next.js, React, TypeScript

**Backend:** FastAPI, Python, async orchestration

**Deployment:** Docker image on AWS App Runner (ECR), Route 53 custom domain, health checks, host allowlist.

**AI Layer:** OpenAI SDK, Google GenAI, DeepSeek, Grok

**Data + Delivery:** SQLite, WeasyPrint, Resend, Clerk

Engineering patterns demonstrated: contract-first outputs, schema validation + retries, provider fallback chains, cache-before-infer, deterministic finance with narrative synthesis (Grounded Finance v2).